

Full Spectrum Inversion of radio occultation signals

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[1] Temperature, pressure, and humidity profiles of the Earth's atmosphere can be derived through the radio occultation technique. This technique is based on the Doppler shift imposed, by the atmosphere, on a signal emitted from a GNSS satellite and received by a low orbiting satellite. The method is very accurate with a temperature accuracy of 1°K, when both frequencies in the GPS system are used. However, difficulties arise when the signal consists of multiple frequencies generated by multipath phenomena in the atmosphere. We demonstrate that, in general, it is possible to determine the arrival times of the different frequency components in a radio occultation signal simply as the derivatives of the phases of the conjugated Fourier spectrum of the entire occultation signal. Based on this property, a novel Full Spectrum Inversion technique for radio occultation sounding capable of disentangling multiple rays in multipath regions is presented. As the entire signal is used in the Fourier transform, a high spatial resolution in the Doppler frequency, and hence in the retrieved temperature, pressure, and humidity profiles, can be achieved. The method is conceptual and computational simple and thus easy to implement. The performance of the Full Spectrum Inversion is demonstrated by applying the technique to simulated signals generated by solving Helmholtz equation with use of the multiple phase-screen technique. Excellent agreement is found between computed bending angle profiles and corresponding solutions to the Abel integral.

INDEX TERMS: 6974 Radio Science: Signal processing; 6904 Radio Science: Atmospheric propagation; 6964 Radio Science: Radio wave propagation; 6969 Radio Science: Remote sensing; **KEYWORDS:** inversion, multipath, radio occultations

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1. Introduction

[2] In general terms, the radio occultation technique is based on the bending of radio waves caused by refractive index gradients in a planetary atmosphere [e.g., Kursinski *et al.*, 1997, 2000]. In Global Navigation Satellite System (GNSS) radio occultations, the bending is measured as radio waves traverse the Earth's atmosphere from a GNSS satellite to a Low Earth Orbit (LEO) satellite (see Figure 1). The atmospheric bending can be retrieved as function of impact parameter from the measured Doppler shifts of the radio signal and the positions and velocities of the satellites. Assuming local spherical symmetry, the measured pairs of bending angle and impact parameter can be inverted through the Abel transform to yield the index of refraction as function of height [Fjeldbo *et al.*, 1971].

[3] In single ray regions, the computation of bending angles is straightforward as they are unambiguously related to the instantaneous frequency of the received signal. Therefore, radio occultation measurements have traditionally been based on the instantaneous frequency. However, radio signals propagating through the lower troposphere may have a very complex structure due to multipath effects caused mainly by water vapor structures [Gorbunov *et al.*, 1996; Gorbunov and Gurvich, 1998; Gorbunov *et al.*, 2000; Sokolovskiy, 2001]. In regions that exhibit multipath, the bending angles cannot be derived directly from the instantaneous frequency of the measured signal because the instantaneous frequency will be related, not to a single pair of bending angle and impact parameter, but to two or more pairs. Another drawback of retrieving the bending angle directly from the instantaneous frequency is that the vertical resolution is limited by the size of the Fresnel zone. Thus, there has been much effort in developing techniques with high vertical resolution that are capable

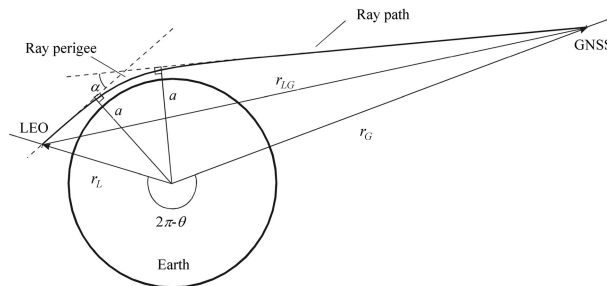


Figure 1. Geometry of GNSS radio occultation. The radio waves propagate through the Earth's atmosphere and the ray paths are bent due to refractivity gradients. The relative motion of the satellites enables the variation of the total bending angle, α , as function of the impact parameter, a , to be derived.

of correctly retrieving the bending angle profile in multipath regions [Gorbunov, 2001].

[4] So far, four high resolution methods have been proposed for processing of radio occultation signals in multipath regions: (1) back-propagation [Gorbunov et al., 1996; Hinson et al., 1997, 1998; Gorbunov and Gurvich, 1998], (2) radio-optics [Lindal et al., 1987; Pavelyev, 1998; Hocke et al., 1999; Sokolovskiy, 2001; Gorbunov, 2001], (3) Fresnel diffraction theory [Marouf et al., 1986; Mortensen and Høeg, 1998; Meincke, 1999], and (4) canonical transform [Gorbunov, 2002a, 2002b]. All of these methods can be termed radio-holographic since they are all based on analysis of the received complex radio signal.

[5] This paper presents a new high-resolution radio-holographic method. The unique in this method is that the instantaneous frequencies are derived directly from a single Fourier transform of the entire complex signal, we therefore refer to this method as Full Spectrum Inversion, abbreviated FSI. For ideal occultations, i.e., occultations with a spherical Earth and perfect circular satellite orbits lying in the same plane, a global Fourier transform can be applied directly to the measured occultation signal. The geometry in real occultations is more complex and the occultation signal and ephemeris data must be resampled with respect to the angle between the radius vectors of the GNSS and the LEO satellites, and frequency variations caused by radial variations in the radius vectors must be removed before the global Fourier transform can be applied. By using the full signal spectrum, the path traversed by the receiving satellite during an occultation can be considered as one synthetic aperture, which yields a very high vertical resolution. The method effectively resolves multiple frequencies in the received signal.

[6] The FSI method is related to the Canonical Transform method: both methods apply some Fourier integral operator (simply Fourier transform for FSI) and use the derivative of the phase after this transformation. For the FSI method, this derivative is related to the

temporal variations of the instantaneous frequency whereas for the Canonical transform this derivative is directly related to the bending angle. For both methods, the amplitudes of the transformed signal describe the distribution of energy with respect to impact parameter. The fundamental difference between the Canonical transform and the FSI method is that the former operates on the spatial distribution of the electromagnetic field along a plane, whereas the latter operates directly on the measured signal along the LEO trajectory.

[7] This paper is organized into the following structure: Section 2 presents the Full Spectrum Inversion technique. Section 3 demonstrates how the technique can be applied to radio occultation signals. For clarity, only ideal occultations are considered. Section 4 covers the application of the Full Spectrum Inversion method in a realistic occultation scenario. In Section 5, we compare bending angle profiles computed using FSI with numerical solutions based on the Abel transform. A brief outline of the use of FSI with real GPS measurements is given in Section 6. The conclusion is presented in Section 7 where essential achievements will be discussed. Besides, the paper is prolonged by two appendixes describing some mathematical details behind radio occultation phase, Doppler shift, and measured amplitude.

2. Full Spectrum Inversion

[8] In this section, we present a general description of Full Spectrum Inversion, considering signals composed of one or more narrow banded subsignal(s) with varying instantaneous frequency. The application of the method to occultation signals will be demonstrated in the following sections. Here we just mention that in a radio occultation context, each subsignal corresponds to a single path and each instantaneous frequency will be related to a given ray.

[9] The instantaneous frequency of a single tone signal can easily be computed as the derivative of the

phase. For signals composed of more than one tone, computation of the instantaneous frequency of each tone is far more complicated. For time-varying signals, a common approach is to use the Short Term Fourier Transform (STFT), which forms the basis for the Radio Optics Method. This transformation splits the signal into smaller blocks and performs the Fourier transform on each block. The length of the signal blocks is chosen as a trade-off between frequency resolution and time resolution. For a detailed discussion of STFT for RO signals, see, for example, *Paveleyev* [1998], *Gorbunov et al.* [2000], and *Sokolovskiy* [2001].

[10] Full Spectrum Inversion provides a simple and efficient tool for deriving the instantaneous frequencies of a signal composed of several narrow banded subsignals. When certain criteria are fulfilled, FSI is capable of resolving the frequency variation of each signal component. Since the FSI is based on the Fourier transform of the entire signal a very high frequency resolution can be achieved compared to the STFT method.

[11] FSI relies on the assumption that the Fourier transform of an entire signal can be computed using the Method of Stationary Phase (MSP) [*Born and Wolf*, 1999]. The MSP method can be applied to a Fourier integral when the following two criteria are fulfilled: (1) For each subsignal, the signal amplitude term is slowly varying compared to the phase term. (2) The second order derivatives of the subsignal phases are large compared to the higher order derivatives. As will be shown below, the use of the FSI method also requires that the representation of the signal in the time-frequency domain yields an unambiguous relation of time as a function of frequency. Consequently, any instantaneous frequency can only occur once, meaning that there is either one or zero stationary phase points in the Fourier integral.

[12] Now, consider a signal composed of several narrow banded subsignals:

$$V(t) = \sum_p Q_p(t) \exp(i\varphi_p(t)), \quad (1)$$

where φ_p is the phase and Q_p the amplitude of the p^{th} subsignal and i is the imaginary unit. The Fourier transform of $V(t)$ yields:

$$\hat{V}(\omega) = \sum_p \int_0^T Q_p(t') \exp[i(\varphi_p(t') - \omega t')] dt', \quad (2)$$

where T is the measurement time.

[13] Consider a given angular frequency for which $\omega = d\varphi_q/dt(t=t_1)$ where q refers to the subsignal containing the angular frequency ω , and t_1 corresponds to the time where the instantaneous frequency of this subsignal is ω . If the requirements listed above are valid, (2) can be evaluated using the MSP method. Assuming

that there is only one stationary phase point, the Fourier integral is dominated by the contribution from the subsignal q that satisfy the relation $\omega = d\varphi_q/dt(t=t_1)$. By using the MSP method, (2) can be approximated as [*Born and Wolf*, 1999]:

$$\hat{V}(\omega) \cong \sqrt{\frac{2\pi i}{d^2\varphi_q/dt^2(t_1)}} Q_q(t_1) \exp[i(\varphi_q(t_1) - \omega t_1)]. \quad (3)$$

Due to the square root term, the phase of the Fourier spectrum will experience a phase shift of $\pi/2$ when $d^2\varphi_q/dt^2$ changes sign.

[14] Equation (3) reveals that we can interpret the different frequency components in the Fourier spectrum as the instantaneous frequencies contained in the signal. Similar results with respect to Fourier spectra and local spatial frequencies can be found in [*Goodman*, 1996].

[15] The arrival times corresponding to the different frequencies in the signal can now be found simply by differentiating the phase, u , of the Fourier transform given by (3):

$$\frac{du}{d\omega} = \frac{d}{d\omega}(\varphi_q(t_1) - \omega t_1) = \frac{d\varphi_q}{dt_1} \frac{dt_1}{d\omega} - \omega \frac{dt_1}{d\omega} - t_1 = -t_1,$$

or on parametric form:

$$(\omega(t), t) = \left(\omega, -\frac{du}{d\omega} \right). \quad (4)$$

which shows that the pairs of instantaneous frequencies and the times where these frequencies occurred are simply given as the Fourier frequencies and the derivatives of the phases of the corresponding conjugated Fourier components.

3. Inversion of Ideal Occultation Data

[16] The Full Spectrum Inversion technique is based on the assumption that the Fourier integral of the entire signal can be evaluated using the MSP method and that any instantaneous frequency can only occur once during an occultation. In this section, we assess the validity of these assumptions. We will assume that the satellites are moving in circular orbits within the occultation plane, that the atmosphere is spherical symmetric, and that geometrical optics gives an adequate description of the signal propagation. The subsignals introduced in the previous section now correspond to different singlepaths and we interpret the radio occultation signal as a sum of several subsignals when multipath occurs.

[17] As shown in Appendix A, the time derivative of the phase of each subsignal (the total Doppler angular frequency) may be expressed as:

$$\omega_q(t) = \frac{d\varphi_q(t)}{dt} = k \left[\frac{dr_G}{dt} \sqrt{1 - \left(\frac{a}{r_G}\right)^2} + \frac{dr_L}{dt} \sqrt{1 - \left(\frac{a}{r_L}\right)^2} + a \frac{d\theta}{dt} \right] \quad (5)$$

in which φ_q is the ray signal phase, k is the wave number, r_L is distance from center of the Earth to the LEO satellite, r_G is distance from center of the Earth to the GNSS satellite, a is the ray impact parameter, and θ denotes the angle between the two radius vectors.

[18] Note that the angular frequency given by (5) is the total Doppler, i.e., the sum of the excess Doppler and geometric Doppler. The latter is defined as the frequency variations caused by the relative movements of the satellites. Under the assumptions described above, the two radial velocity components are zero and the angular velocity, $d\theta/dt$, is constant, implying that:

$$\omega_q(t) = \frac{d\varphi_q(t)}{dt} = k a \frac{d\theta}{dt}. \quad (6)$$

That is, the total frequency of each ray is proportional to the ray impact parameter, ensuring a unique relation between ray impact parameter and the ray Doppler frequency. In agreement with *Gorbunov* [2001], it is assumed that rays are uniquely identified by their impact parameter. Therefore, we conclude that, under the given assumptions, any instantaneous frequency will occur only once during an occultation.

[19] From geometrical optics, it can be shown that the amplitude Q of a radio occultation (sub)-signal may be expressed as (see Appendix B):

$$Q = \left[\frac{P}{2\pi} \frac{a}{(r_G r_L)^2 \sin(\theta) \sqrt{1 - \left(\frac{a}{r_G}\right)^2} \sqrt{\left(\frac{a}{r_L}\right)^2 \frac{d\theta}{da}}} \right]^{1/2}, \quad (7)$$

in agreement with S. S. Leroy (Amplitude of an occultation signal in three dimensions, submitted to *Radio Science*, 2001), where P is the transmitted signal power. In single path regions (7) represents the total signal amplitude, whereas in multipath regions (7) yields the amplitude of each subsignal. It follows from (6) that:

$$\frac{d\theta}{da} = k \left(\frac{d\theta}{dt} \right)^2 \left(\frac{d^2\varphi_q}{dt^2} \right)^{-1}. \quad (8)$$

Inserting (6) and (8) into (7) yields:

$$Q_q = \frac{\left[\frac{P \frac{d^2\varphi_q}{dt^2} \left(\frac{\omega_q}{k} \right)}{2\pi k \sin(\theta(\omega_q)) \left(\frac{d\theta}{dt} r_L \right) \left(\frac{d\theta}{dt} r_G \right)} \right]^{1/2}}{\left[\left(\left(\frac{d\theta}{dt} r_L \right)^2 - \left(\frac{\omega_q}{k} \right)^2 \right) \left(\left(\frac{d\theta}{dt} r_G \right)^2 - \left(\frac{\omega_q}{k} \right)^2 \right) \right]^{1/4}}. \quad (9)$$

Since the relative variations in ω_q , r_L , and r_G are generally small compared to the variations in $d^2\varphi_q/dt^2$, we can assume that Q_q is proportional to $(d^2\varphi_q/dt^2)^{1/2}$.

[20] If the higher order phase terms are small compared to the second order phase term, the phase variations near a stationary phase point are dominated by the second order derivative of the phase, whereas the amplitude variations are dominated by the first order derivative of the amplitude. Now, if we require that the relative variations in the amplitude shall be small during a time interval where the phase is increased or decreased with a factor of π then the following must be true:

$$\left| \frac{d^3\varphi_q}{dt^3} \right| \ll \sqrt{2\pi} \left| \frac{d^2\varphi_q}{dt^2} \right|^{3/2}, \quad (10)$$

where it has been assumed that the subsignal amplitude is proportional to $(d^2\varphi_q/dt^2)^{1/2}$ as implied by (9). Equation (10) reveals that the relative amplitude variations are always small when the higher order phase derivatives are negligible compared to the second order derivative. Hence, for the assumptions stated in the beginning of this section, the only requirement for the validity of the FSI method is that the higher order phase terms of each subsignal must be small compared to the second order or higher order phase terms. There are two obvious cases where this requirement may be violated; when $d^2\varphi_q/dt^2 = 0$, and when the instantaneous frequency varies rapidly. General assessment of the impact on the FSI resolution caused by higher order phase derivatives is difficult. Instead, the validity of the method should be demonstrated through simulations using realistic refractivity fields. However, this is a subject of a future study and in the following it will be assumed that the contribution from higher order phase terms can be neglected.

[21] Now, inserting (9) into (3) yields the Fourier transform of the occultation signal:

$$\begin{aligned} \hat{V}(\omega) &\equiv \sqrt{2\pi i} \left(\frac{d^2\varphi_q}{dt^2} \right)^{-1/2} Q_q(t_1) \exp[i(\varphi_q(t_1) - \omega t_1)] \\ &= U \exp(iu(\omega)), \end{aligned}$$

in which

U

$$= \left(\frac{P \left(\frac{\omega}{k} \right) \left(\frac{d\theta}{dt} r_L \right)^{-1} \left(\frac{d\theta}{dt} r_G \right)^{-1}}{k \sin(\theta(\omega)) \sqrt{\left(\frac{d\theta}{dt} r_L \right)^2 - \left(\frac{\omega}{k} \right)^2} \sqrt{\left(\frac{d\theta}{dt} r_G \right)^2 - \left(\frac{\omega}{k} \right)^2}} \right)^{1/2} \quad (11)$$

and

$$u(\omega) = \exp \left(i \left(\varphi_q(t_1) - \omega t_1 + \frac{\pi}{4} \right) \right), \quad (12)$$

where all derivatives are evaluated for $t = t_1$.

[22] From (12) it follows that the relative variations in the magnitude of $U(\omega)$ is small and the power spectrum of an entire radio occultation signal will therefore have an approximately rectangular shape if the assumptions stated in beginning of this section are fulfilled. This is an important property because it allows us to use the shape of the signal spectrum to check whether our assumptions about the occultation signal are valid.

4. Inversion of Realistic Occultation Data

[23] The preceding section demonstrated the application of FSI to radio occultation signals under the assumption that the satellites were orbiting a spherical Earth in the same plane following perfect circular orbits. In this section, we show how FSI can be applied in a similar manner for realistic occultations. In this context, we define realistic occultations as occultations where the Earth is oblate and the satellite orbits are only nearly circular (like those used by the GPS satellites, the GPS/MET mission, and the CHAMP satellite) including also occultations where the GNSS and the LEO orbits are not in the same plane.

[24] When the oblateness of the Earth is taken into account, inversion of the occultation data should be performed assuming local spherical symmetry tangential to the Earth ellipsoid [Syndergaard, 1998]. Generally, the center of local curvature does not coincide with the center of the satellite orbits. Consequently, we cannot assume that the satellite orbits are circular when inverting the occultation data, as radial velocities and accelerations are introduced when the satellite positions are described relative to the center of local curvature. Furthermore, when the angle between the two satellite orbits are not zero, there is no longer a linear relation between time and θ .

[25] The presence of higher order derivatives of θ and the presence of radial accelerations may ruin the unambiguous relation of time as function frequency. In these cases, FSI will give very noisy results in the

affected frequency band. The noise/oscillations in the retrieved arrival times represent an interference problem, as more arrival times correspond to the same instantaneous frequency. These oscillations have the same nature as the oscillation in the instantaneous frequency observed in multipath regions. Hence, a unique relationship between ray frequency and ray impact parameter cannot be assumed in general, as in the previous section. When interference occurs, direct application of (5) results in a noisy bending angle profile for the affected impact parameters. Consequently, we must eliminate the interference caused by the presence of higher derivatives in both radius vectors and θ prior to the application of FSI.

[26] One way to eliminate this kind of interference is to use a modified version of the back-propagation technique where we instead of back-propagating the field to a plane propagates the field to a circular LEO orbit approximating the real LEO orbit. However, here we will introduce another approach, which offers a simpler implementation and requires far less computer power. The idea is to remove the influence from radial accelerations and higher order derivatives of θ in two simple steps.

[27] First, it is straightforward to account for the influence of the higher order derivatives of θ . If we instead of differentiating the signal phase with respect to time differentiate with respect to θ then (5) can be rewritten as:

$$\frac{d\varphi_q}{d\theta} = ka + k \frac{dr_L}{d\theta} \sqrt{1 - \left(\frac{a}{r_L} \right)^2} + k \frac{dr_G}{d\theta} \sqrt{1 - \left(\frac{a}{r_G} \right)^2}, \quad (13)$$

where $d\varphi_q/d\theta$ is a pseudo angular frequency, which is approximately proportional to $d\varphi_q/dt$. Equation (13) shows that the contributions from higher order derivatives in θ can be eliminated if we instead of time use θ as the independent variable.

[28] Second, interference caused by radial variations can be removed by correcting the signal phase by a phase factor representing the frequency variation introduced by the last two terms in (13). This, of course, requires a priori knowledge of a or a sufficient accurate model of $a(\theta)$. An obvious choice of model is the impact parameter profile derived from the smoothed signal phase. Assuming that the maximum difference between the different impact parameters in a multipath zone is of the order of 10 km, this model yields a relative model impact parameter accuracy of the order of $O(10^{-3})$ or better in multipath regions, whereas the model is nearly perfect in single path regions.

[29] From the previous discussions, we conclude that FSI is suitable for inverting real occultation signals if we instead of time use θ as the independent variable and remove the effects of radial variations in the satellite

position vectors. In practice, the variable transformation is most easily performed by resampling the original signal and ephemeris data with respect to θ . Radial variations can be removed by imposing an appropriate frequency shift through the following operation:

$$V_s(\theta) = V(\theta)\exp(-i\phi_m(\theta)), \quad (14)$$

where

$$\phi_m(\theta) = k \int_{\theta_0}^{\theta} \frac{dr_L}{d\theta} \sqrt{1 - \left(\frac{a_m}{r_L}\right)^2} + \frac{dr_G}{d\theta} \sqrt{1 - \left(\frac{a_m}{r_G}\right)^2} d\theta', \quad (15)$$

in which θ_0 is the initial angle between the radius vectors, $V(\theta)$ represents the measured signal resampled with respect to θ , $V_s(\theta)$ represents the measured signal after removal of the frequency shift caused by radial variations, and a_m represents a model of the true impact parameter variation.

[30] The pseudo frequency variation $d\phi_s/dt$ of V_s can be retrieved using FSI, subsequently the impact parameter variation can be derived from (13) using:

$$\frac{d\phi_q}{d\theta} = \frac{d\phi_s}{d\theta} + \frac{d\phi_m}{d\theta}. \quad (16)$$

Finally, the corresponding bending angles, α , can be derived from the geometrical relations [e.g., *Kursinski et al.*, 2000]:

$$\alpha = \theta + \phi_G + \phi_L - \pi, \quad (17)$$

and

$$a = r_L \sin(\phi_L) = r_G \sin(\phi_G), \quad (18)$$

where ϕ_G and ϕ_L are the angles between the ray path and the satellite radius vector at the GNSS and the LEO satellites, respectively.

5. FSI of Simulated Occultation Signals

[31] In this section, we validate the Full Spectrum Inversion technique. We have chosen to validate the method using an atmosphere without horizontal variations described by known model refractivity profiles. We compare model bending angle profiles, which have been computed from the refractivity profiles (via Abel transformation), with bending angle profiles computed by FSI from simulated radio occultation signals. These signals are generated by applying a wave propagator to atmospheres with refractivity distributions described by the model refractive profiles. Two different test cases have been investigated. In the first case, we consider ideal occultations and an atmosphere with multipath. In the second case, we consider realistic occultations and an atmosphere with strong multipath.

[32] Our simulations are based on solutions to the parabolic approximation of the Helmholtz wave equation computed using the split-step algorithm, also known as the multiple phase-screen technique [*Gorbunov and Gurvich*, 1998; *Sokolovskiy*, 2001]. This approach offers a forward propagating model, capable of simulating the propagation of radio waves for most realistic atmospheric conditions. There are two primary limitations to this technique: (1) The backscattered field is neglected. (2) Accurate calculations are restricted to near-horizontal propagation. The propagation from the last screen to the LEO orbit is calculated using the Fresnel diffraction integral.

[33] All the simulations have been performed using a wave optics propagator developed by M. E. Gorbunov. In the simulations, the number of phase screens was set to 1200 and a sampling frequency of 70 Hz was used.

[34] In the first test case, the model refractivity profile N is given as function of height, h , and can be described by the following equation:

$$N(h) = 315 \exp\left(\frac{-h}{7.35\text{km}}\right) + B \exp\left(\frac{-(h - BH)^2}{0.05\text{km}^2}\right). \quad (19)$$

[35] This refractivity profile represents an exponential with a Gaussian shaped bump. In (19) BH is the height of the bump, and B determines the size of this bump. The first part of the equation is the refractivity of the reference atmosphere defined by the *International Telecommunications Union (ITU)* [1981]. The second part of the equation determines the strength and height of the refractivity bump. This bump in the refractivity profile can be used to simulate multipath effects. The duration of a possible multipath situation is determined by the magnitude of the parameter B . In the simulations, B and BH have been set to 20 and 3 km, respectively, which yields an atmosphere with significant gradients leading to multipath propagation. At this point, the satellites are moving in circular orbits within the occultation plane.

[36] In Figure 2 the amplitude of the occultation signal is depicted. Signs of strong multipath interference are noticed from 21 to 33 s as significant fluctuations in the signal amplitude. The simulated signal is amplitude and excess phase sampled like real radio occultations signals. Before a signal can be processed using the FSI method, the total signal phase has to be reconstructed in order to restore the 'original' signal. This is done by adding the phase corresponding to the geometrical Doppler, followed by up-sampling of the signal through interpolation of the amplitude and the new phase. When the signal is interpolated, the sampling rate must correspond to at least twice the new signal bandwidth in order to satisfy the sampling theorem.

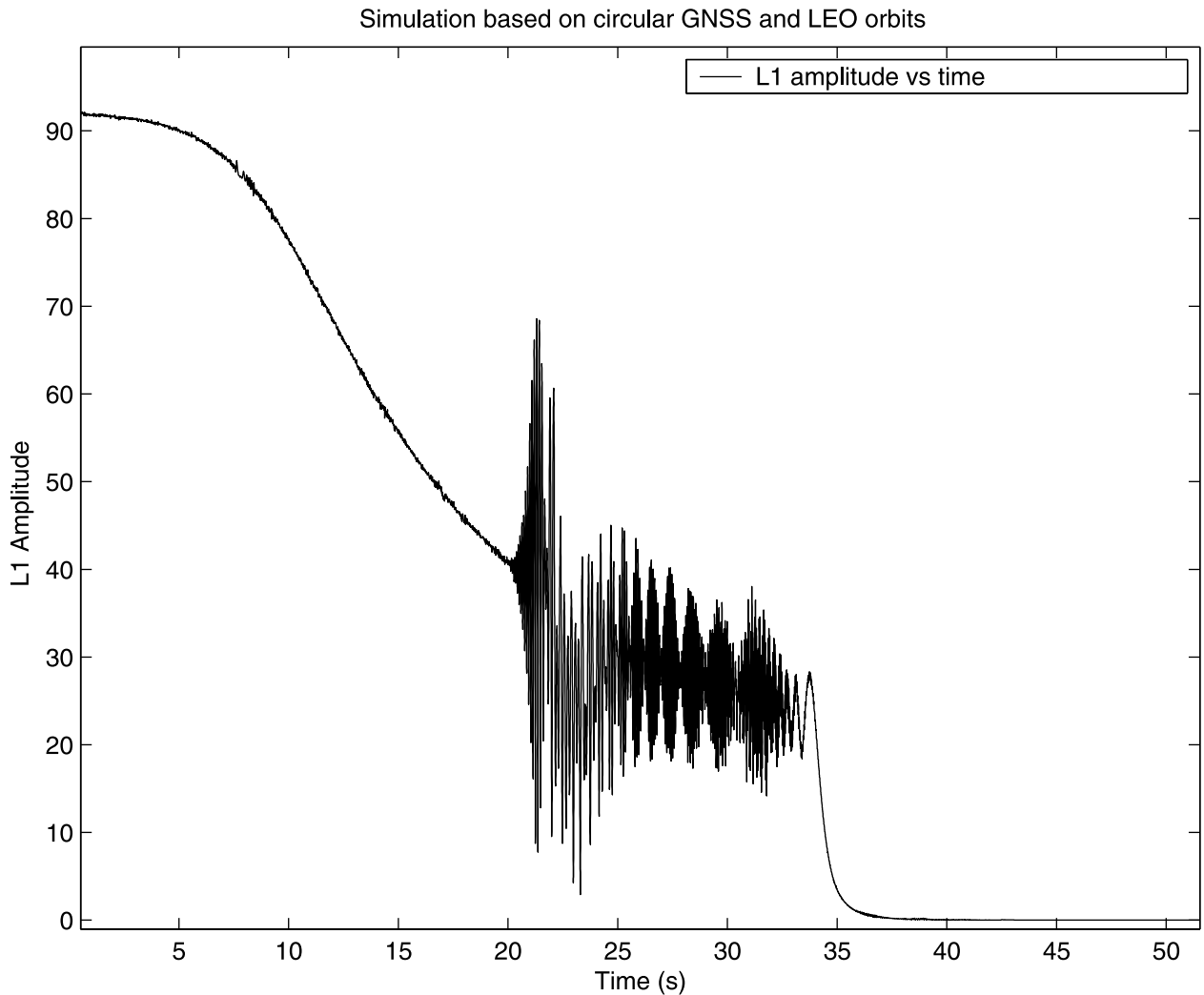


Figure 2. Signal amplitude variations corresponding to simulation with ideal satellite orbits. The amplitude fluctuations in the interval from 21 to 33 s are caused by multipath.

[37] For the current simulation, the center frequency was removed and the signal was up-sampled by a factor of 11. After retrieval of the instantaneous frequency variation, the center frequency was added to the retrieved Doppler frequencies before the bending angle profile was computed.

[38] The results from a FFT of the restored signal are shown in Figure 3. Within the bandwidth of the signal, the FFT spectrum is flat as predicted by (9). Yet, a dip in the spectrum is observed at a frequency of -60 Hz caused by the multipath. The phase of the conjugated signal spectrum can be differentiated with respect to frequency to derive the frequency variation (see (4)). By doing this, the frequency variation depicted in Figure 4 is obtained. The figure shows that FSI is capable of

resolving the frequency variations of all the three sub-signals in the occultation signal showing the expected S-signature of a multipath zone. For comparison, Figure 4 also depicts the instantaneous frequency computed as the time derivative of the signal phase. It is seen that the two curves are overlapping except in the area with multipath effects, which is on the abscissa from 21 to 33 s. In this interval, the instantaneous frequency fails to resolve the different subsignals; instead, a very irregular Doppler variation is erroneously predicted. This is to be expected, since pure phase detection cannot reveal multipath.

[39] In Figure 5 we compare the bending angle profile computed from the frequency variation predicted by FSI, with the corresponding bending angle profile

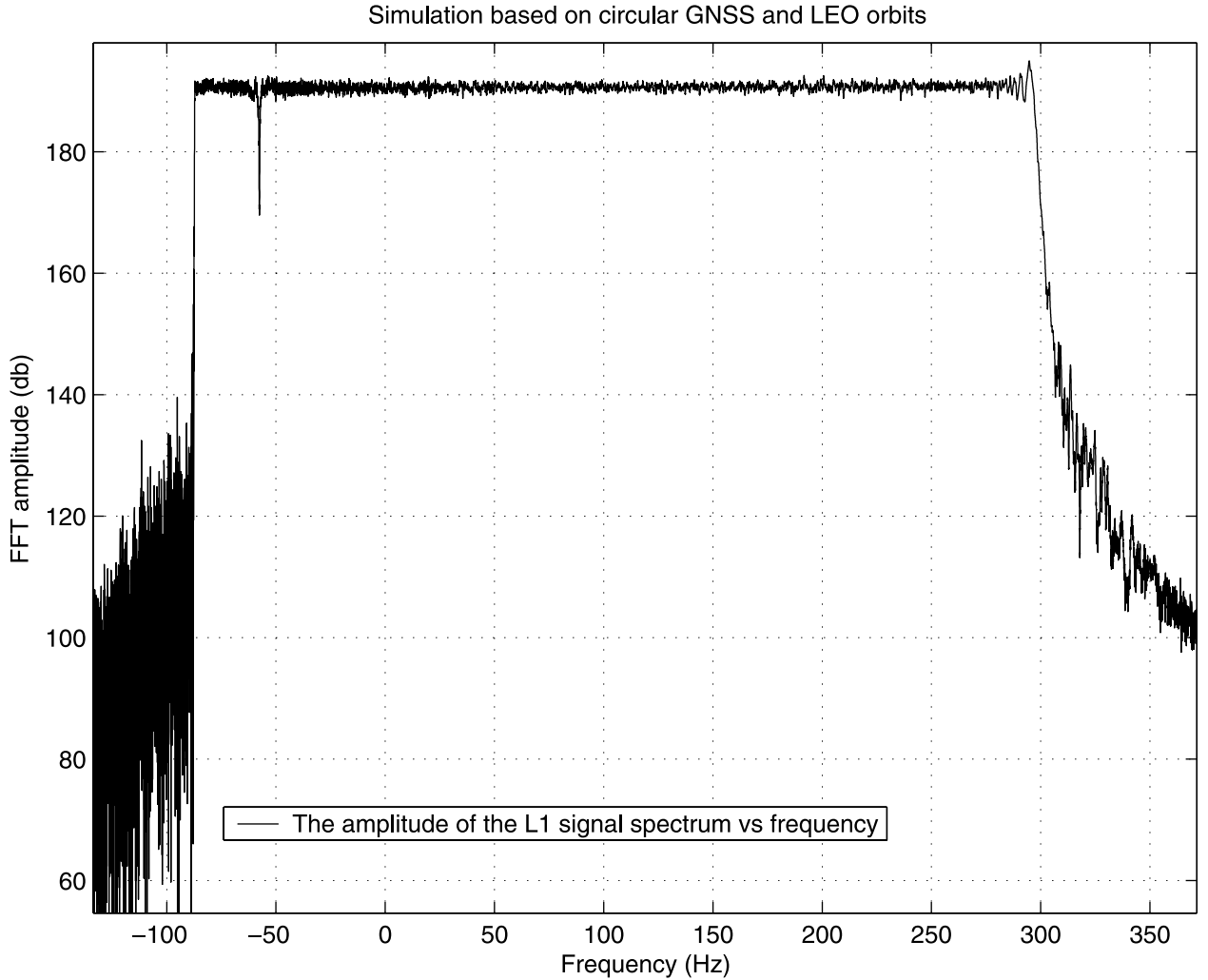


Figure 3. Power spectrum of up-sampled occultation signal corresponding to simulation with ideal satellite orbits. The small dip at -60 Hz is due to multipath.

obtained as the Abel transform of the model refractivity profile. We see an excellent agreement between the two curves demonstrating FSI's ability to retrieve bending angle profiles in regions with strong multipath. We also notice that the small deviations from the flat power spectrum observed in Figure 3 do not seem to have any appreciable effect on the retrieval of the bending angle profile.

[40] We will now apply the FSI method to a simulation with realistic orbits, an oblate Earth, and an atmosphere with strong multipath. The model refractivity profile, N_B , is at this point given as function of height, h , by the following equations:

$$N_A(h) = 400\exp(-h/8\text{km}), \quad (20a)$$

$$N_B(h) = N_A(h) \left[1 - 0.05 \left(\frac{2}{\pi} \right) \tan^{-1} \left(\frac{h - 7\text{km}}{0.05\text{km}} \right) \right], \quad (20b)$$

which are similar to the refractivity model used by Sokolovskiy [2001]. Equation (20) describes an exponential profile that is displaced at a height of 7 km. In the simulated occultation, the LEO orbit corresponds to the GPS/MET orbit and the GNSS orbit corresponds to a typical GPS orbit. The average antenna azimuth angle was 27 degrees for the occultation. The amplitude variation of the simulated radio occultation signal is plotted in Figure 6. Between 27 and 35 s, the signal amplitude shows significant oscillations due to multipath interference.

[41] Following the description in Section 4, the simulated signal and the corresponding ephemeris data are

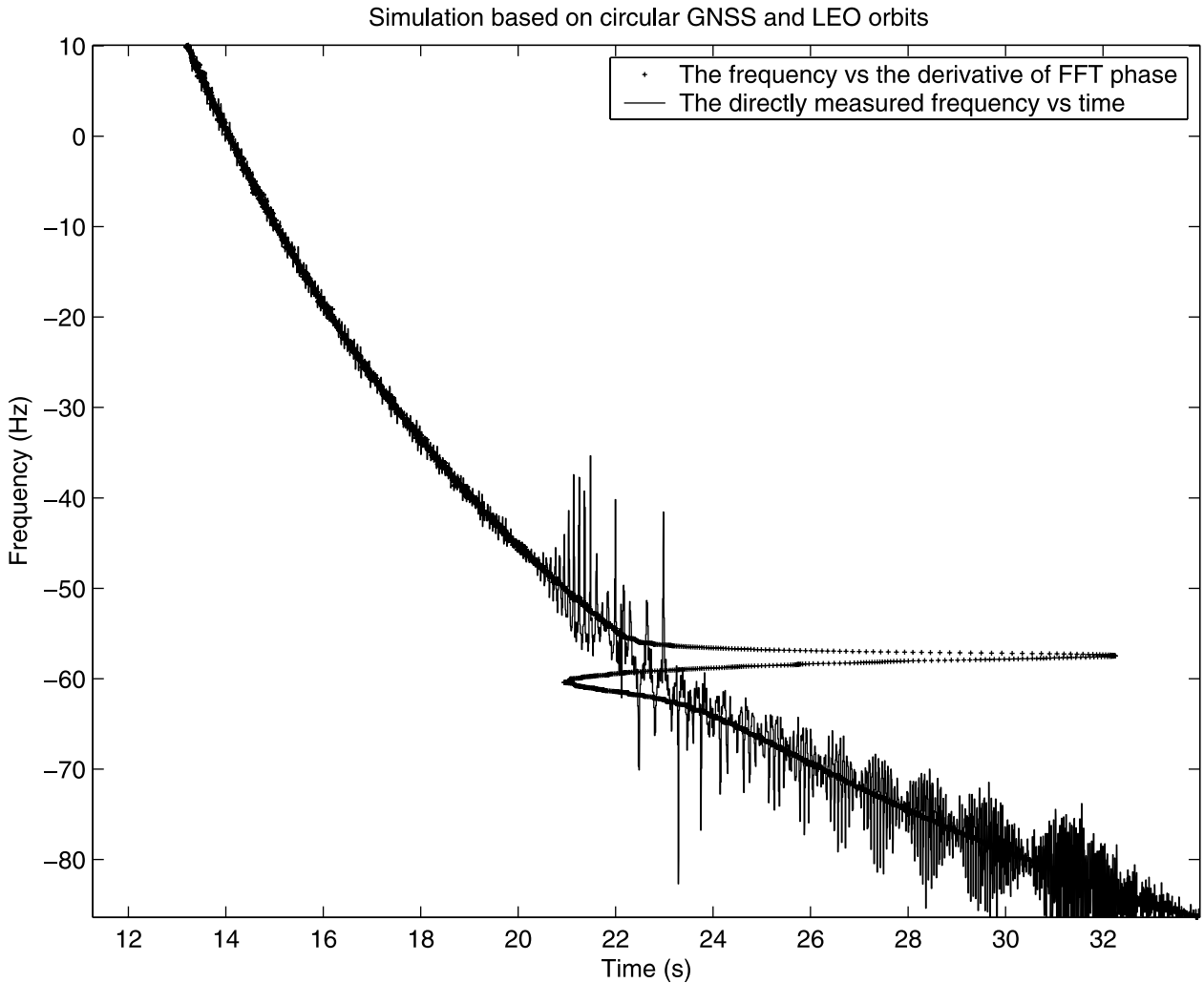


Figure 4. Frequency versus time reconstructed by FSI (curve of small crosses) and by differentiation of the signal phase (solid curve) for ideal orbits. It is readily noticed that the FSI method is capable of resolving multipath.

resampled with respect to the angle between the two radius vectors and frequency variation caused by radial variations in the satellite positions are removed. Subsequently, we apply the same processing scheme as the one used above. The power spectrum of the complex up-sampled signal is depicted in Figure 7. We notice that the spectrum, as in the previous ideal case, is quite flat. A comparison between the bending angle profile computed using FSI and a corresponding bending angle profile computed by means of an Abel transformation of the refractivity profile given by (20) is presented in Figure 8. We observe a very fine agreement between the FSI profile and the corresponding solution to the Abel transform even in the multipath region.

[42] Close inspection of Figures 4, 5, and 7 reveals small oscillations in the frequency variation and in the bending angle profiles. These oscillations are not related to variations in the refractivity index, but are results of the finite window used in the Fourier transform.

6. Application to Real Data

[43] The optimal application of the FSI method to real radio occultation data is a subject of a future study. In this section, we just briefly discuss two important issues related to processing of measured occultation data; ionospheric calibration and retrieval in the multipath free upper atmosphere.

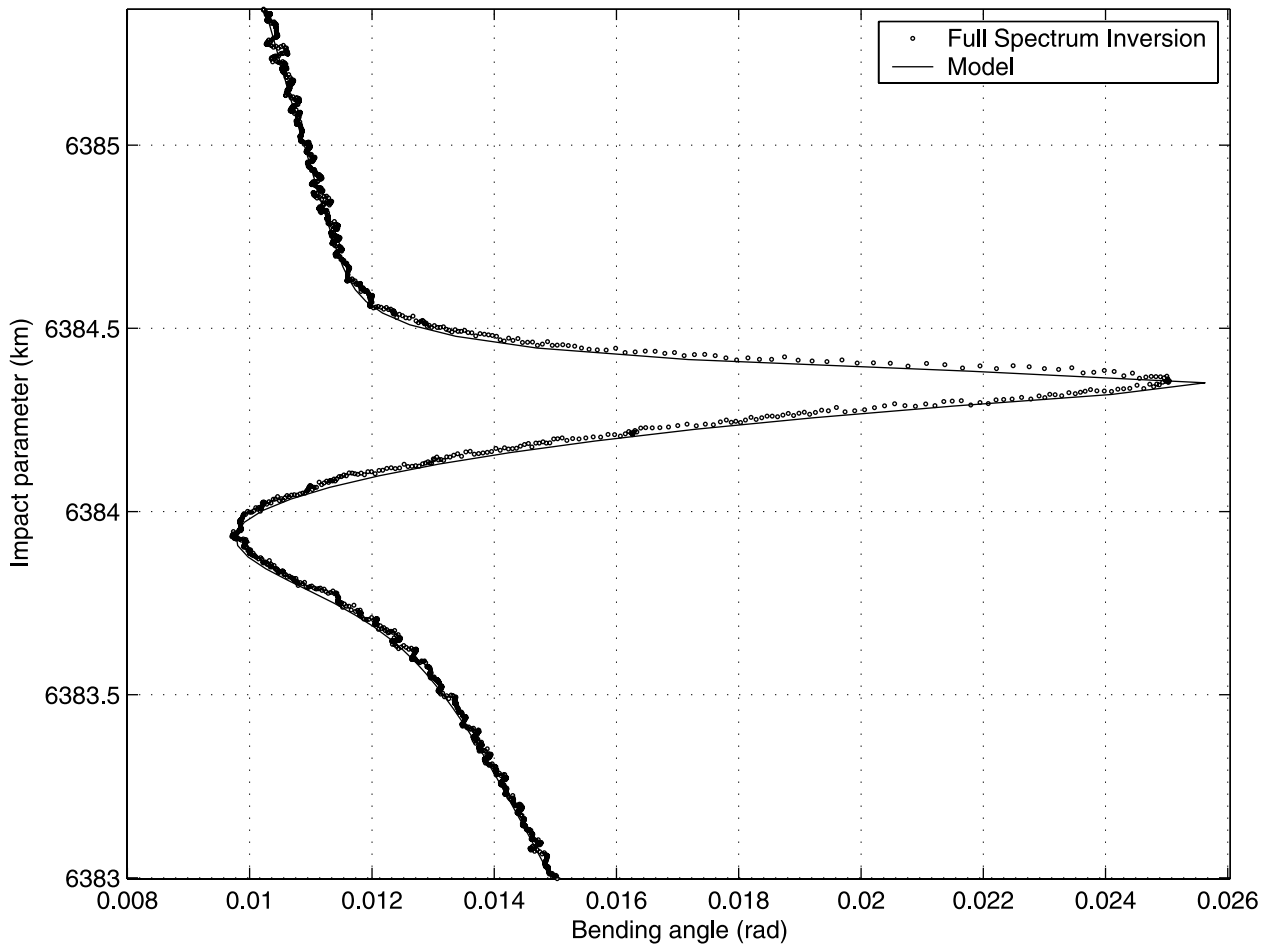


Figure 5. Bending angle versus impact parameter reconstructed by FSI (curve of small circles) and derived directly from the Abel transform (solid line) for ideal orbits.

[44] When applying FSI to real GPS occultations, the ionospheric calibration can be done by either correcting the phase or the bending angle [Syndergaard, 2000]. However, it is desirable to apply the ionospheric calibration on the bending angle, since calibration on the phase may introduce phase accelerations that can introduce interference phenomena in the derived arrival times as described in Section 4.

[45] Above the tropopause, the occurrence of multipath is rare and it might therefore be desirable to use traditional processing technique and derive the instantaneous frequency directly from the measured phase. The advantage of doing this is that the small oscillations in the bending angle profiles caused by the final length of the Fourier transform, described earlier, can be avoided. However, it must be noted that it is always advantageous to perform Fourier transformation on the entire signal in order to use the largest possible window even if FSI is used only in the lower atmosphere.

7. Summary and Conclusion

[46] A new radio holographic inversion method for GNSS-LEO radio occultations has been presented, which we refer to as the Full Spectrum Inversion method. This technique yields sub-Fresnel vertical resolution and is capable of disentangling multiple rays in multipath regions, which are known to appear quite often in the lower troposphere.

[47] Full Spectrum Inversion is based on the fact that when a signal is composed of several narrow-banded subsignals the arrival time of the frequency components in the Fourier spectrum is simply given as the derivative of the phases with respect to the frequency of the conjugated Fourier components. The method is valid, if the following three criteria are fulfilled: (1) Any instantaneous frequency can occur only once. (2) For each subsignal, the signal amplitude term must be slowly varying compared to the phase term. (3) The second

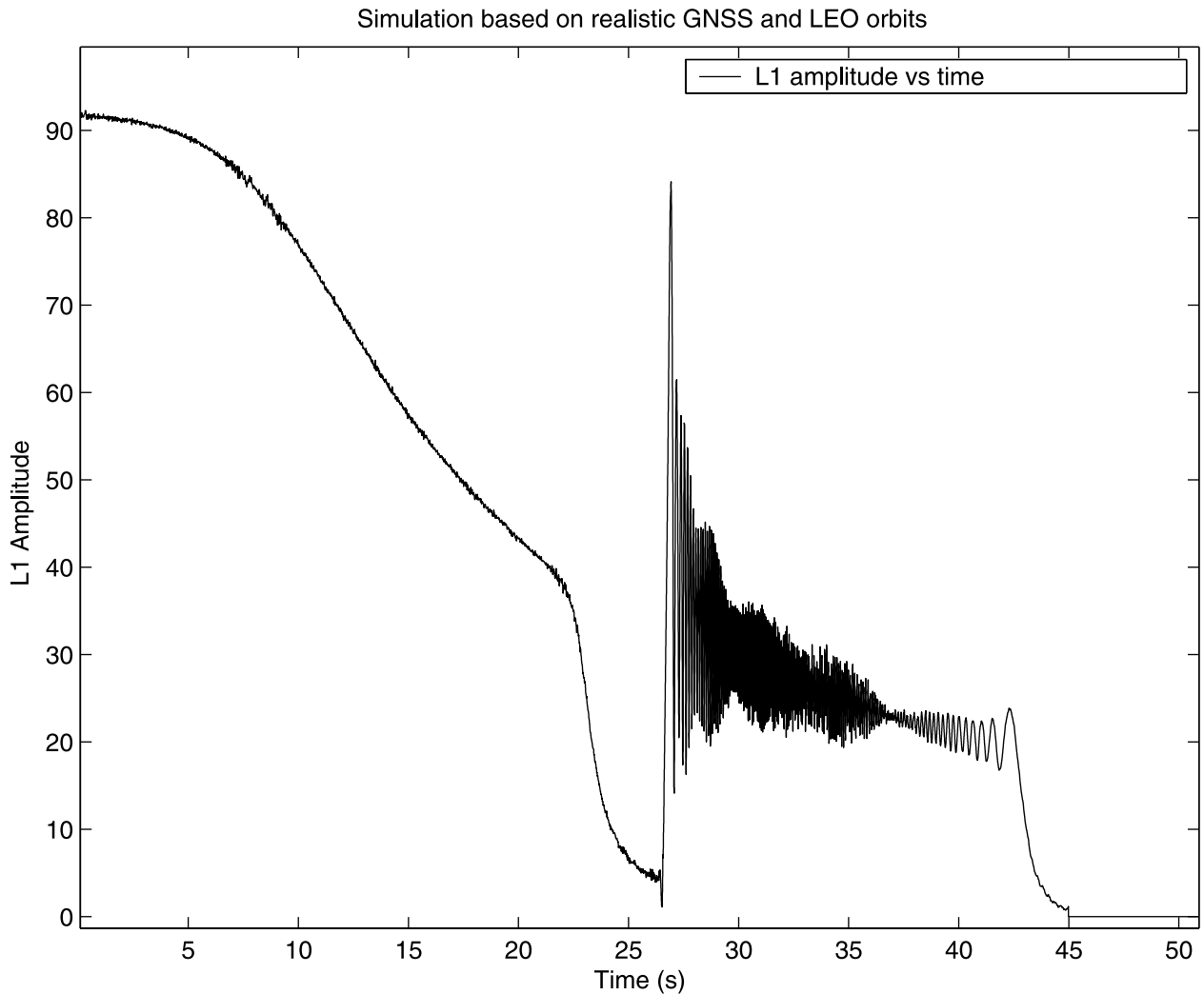


Figure 6. Amplitude variations corresponding to simulation with realistic satellite orbits.

order derivatives of the subsignal phases must be large compared to the higher order derivatives.

[48] When using Full Spectrum Inversion to invert radio occultation data, a distinction between ideal occultations and realistic occultations must be made. Here ideal occultations are defined as occultations with a spherical Earth and perfect circular orbits lying in the same plane. On the other hand, realistic occultations are defined as occultations with an oblate Earth and approximately circular orbits lying in two different planes. In the former case, a global Fourier transform can be applied directly to the measured signal. Realistic occultations require that the occultation signal and ephemeris data are resampled with respect to the angle between the radius vectors of the GNSS and the LEO satellites, and that frequency variations caused by radial variations in the radius vectors are removed before a global Fourier

transform can be applied. Removal of those unwanted frequency variations requires a priori knowledge of the variations in impact parameter during an occultation, which can be estimated, e.g., from the smoothed signal phase.

[49] The performance of Full Spectrum Inversion has been verified by applying the technique to simulated signals generated by solving Helmholtz equation with the multiple phase-screen technique. Two different test cases were considered. In the first case, ideal satellite orbits and a perfect spherical atmosphere with multipath was considered, while in the second case, realistic orbits and an oblate atmosphere with strong multipath was investigated. From the simulated occultation signals, the corresponding bending angle profiles were reconstructed using Full Spectrum Inversion. Comparison between the reconstructed bend-

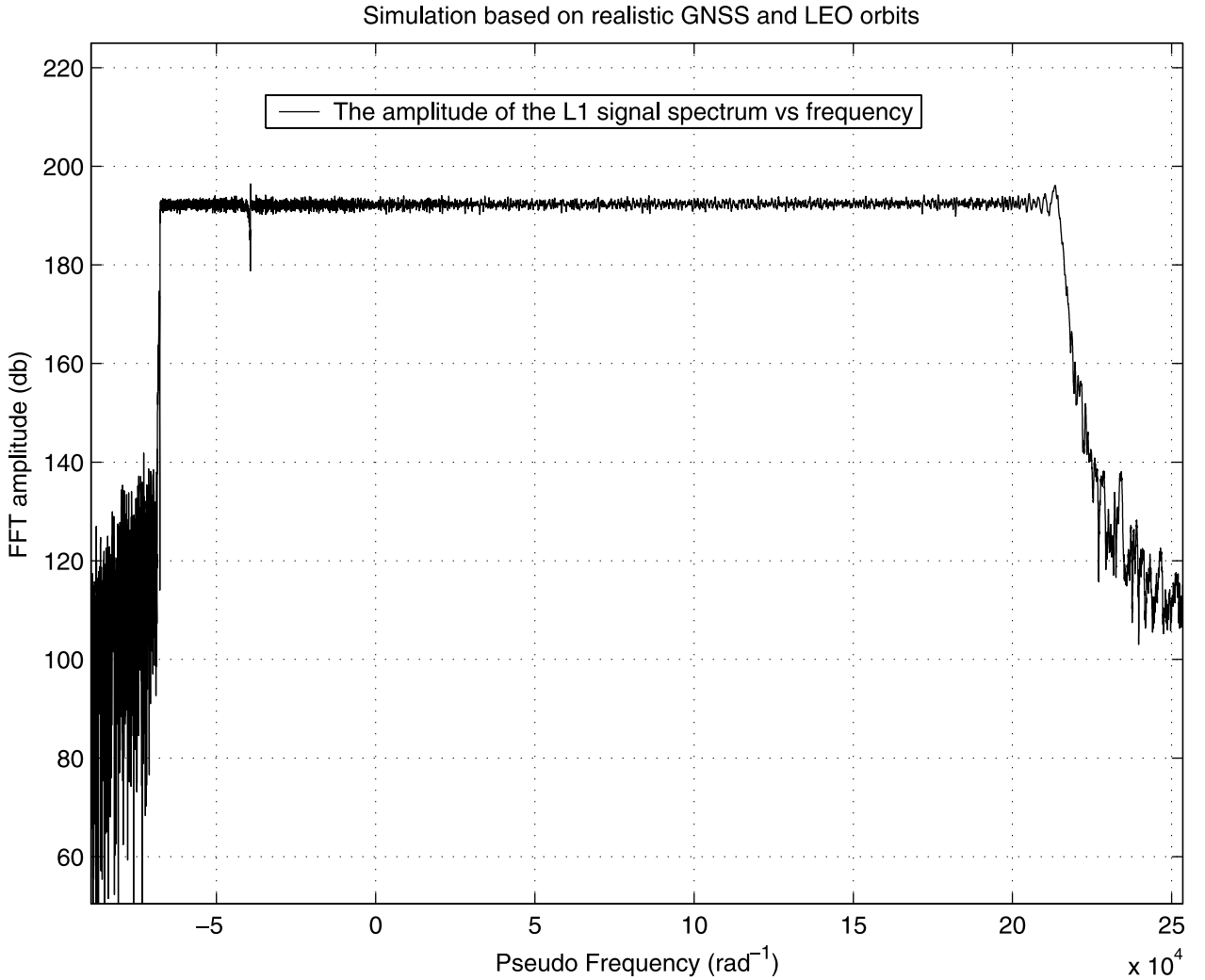


Figure 7. Signal power spectrum corresponding to simulations with realistic satellite orbits

ing angle profiles and bending angle profiles obtained directly from the refractivity profiles through the Abel transformation shows excellent agreement demonstrating the ability of the Full Spectrum Inversion method to retrieve the correct bending angle profile in multipath regions.

Appendix A: The Geometrical Optics Radio Occultation Phase and Doppler

[50] The phase φ , of a ray propagating along a path S , from S_G to S_L is given by

$$\varphi = k \int_{S_G}^{S_L} n ds, \quad (\text{A1})$$

where n is the index of refraction along the path and k is the wave number. The path is determined by the condition that the integral (A1) is stationary along the path; this is due to the principle of Fermat. If the index of refraction in the atmosphere can be assumed to vary only in the radial direction of the Earth, (A1) can be expressed in explicit terms. With a spherical symmetric atmosphere the Bouger's formula for the media can be used, which yields:

$$n r \sin \psi = a, \quad (\text{A2})$$

where r is the radial distance from the Earth, Ψ is the angle between the tangent to the propagation path and the radius vector from the center of the Earth, and a is the impact parameter. The impact parameter is constant along the ray path, i.e., $da/ds = 0$. From the geometry of

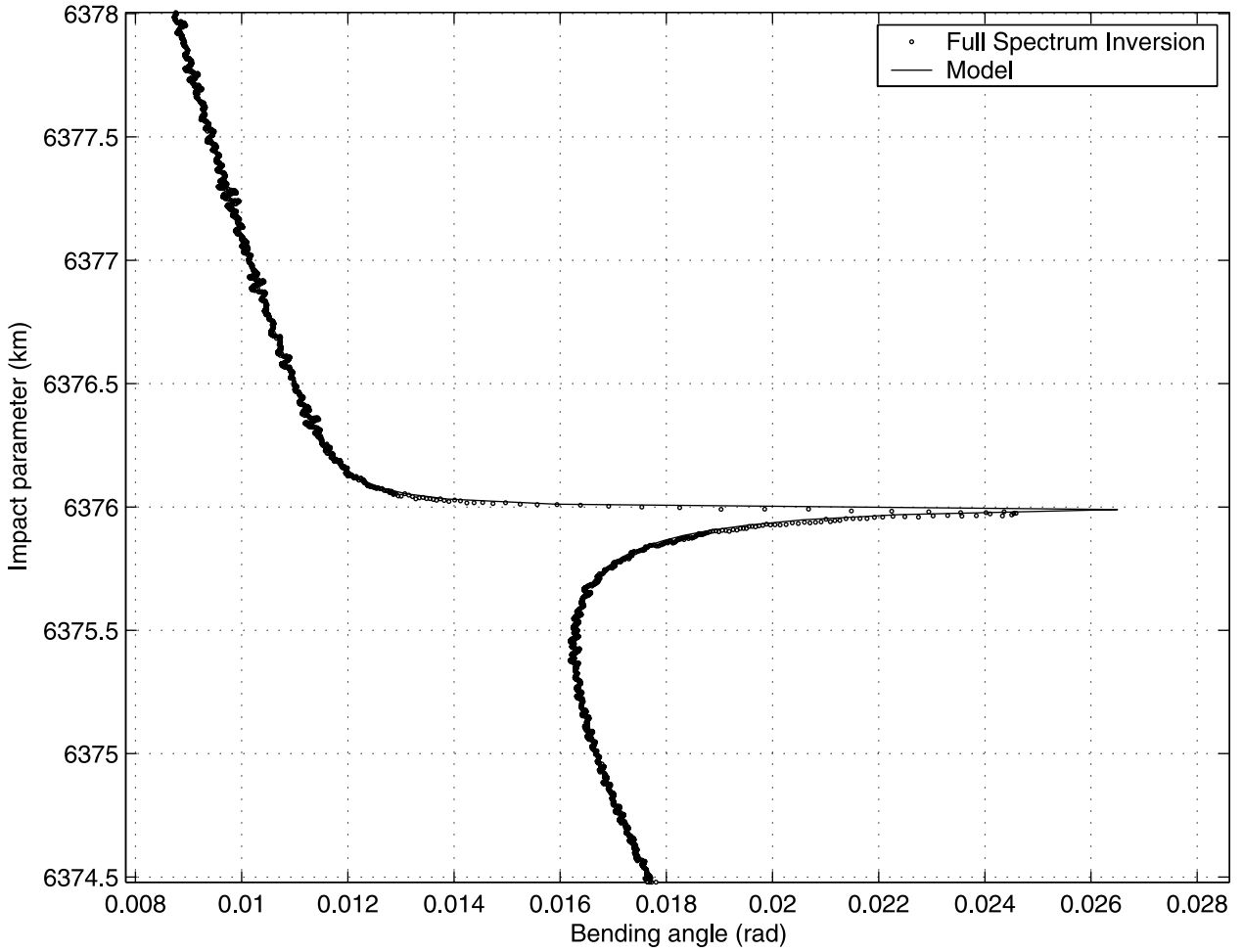


Figure 8. Bending angle versus impact parameter reconstructed by FSI (curve of small circles) and derived directly from the Abel transform (solid line) for realistic satellite orbits.

the occultation it can be shown that

$$\sin(\psi) = \frac{r \left(\frac{d\theta}{dr} \right)}{\sqrt{1 + r^2 \left(\frac{d\theta}{dr} \right)^2}} \quad ds = \sqrt{1 + r^2 \left(\frac{d\theta}{dr} \right)^2} dr. \quad (\text{A3})$$

Here (r, θ) is the polar coordinates of a point on the ray. Using the kinematic relations stated in (A3) and Bouguer's formula, the optical path between the transmitter and the receiver becomes:

$$\begin{aligned} \varphi &= k \int_{s_G}^{s_L} n ds \\ &= k \int_{r_0}^{r_L} n \frac{nr dr}{\sqrt{n^2 r^2 - a^2}} + k \int_{r_0}^{r_G} n \frac{nr dr}{\sqrt{n^2 r^2 - a^2}}, \end{aligned} \quad (\text{A4})$$

where r_G and r_L are the distances from the GNSS and LEO satellites to the center of the Earth, respectively. r_0 is the distance to the tangent point (i.e., $\Psi = \pi/2$) from the Earth center. In the tangent point r_0 is given by $n(r_0) r_0 = a$. Equation (A4) can be rewritten in several forms. The one used here yields:

$$\begin{aligned} \varphi &= k \int_{r_0}^{r_L} \frac{1}{r} \sqrt{n^2 r^2 - a^2} dr \\ &+ k \int_{r_0}^{r_G} \frac{1}{r} \sqrt{n^2 r^2 - a^2} dr + k a \theta. \end{aligned} \quad (\text{A5})$$

[51] The instantaneous Doppler frequency can be defined as the time derivative of the phase. In (A5) the quantities r_L , r_G , r_0 , a , and θ are all functions of time. Differentiating (A5) with respect to time gives the ray Doppler frequency $\omega(t)$:

$$\omega(t) = \frac{d\varphi}{dt} = k \frac{\sqrt{r_G^2 - a^2}}{r_G} \frac{dr_G}{dt} + k \frac{\sqrt{r_L^2 - a^2}}{r_L} \frac{dr_L}{dt} + ka \frac{d\theta}{dt}. \quad (\text{A6})$$

This expression for the Doppler frequency is only one among many others found in the literature. It has a very simple physical meaning especially when the radial velocities are zero or very small. In this case, the Doppler frequency is simply proportional to the impact parameter.

Appendix B: Radio Occultation Amplitude

[52] In order to determine the amplitude of a radio signal, the intensity of each ray must be known. *Born and Wolf* [1999, chapter 3.1.2] express the ratio between the intensities, I_1 and I_2 , in two points of a ray as:

$$\frac{I_2}{I_1} = \frac{n_2}{n_1} e^{-\int_{s_1}^{s_2} \frac{\nabla^2 S}{n}}, \quad (\text{B1})$$

where $S=S(r, t)$ is the eikonal that describes the optical wave surfaces or the geometrical wave fronts and $n=n(r)$ is the index of refraction of the medium. Initial and final quantities are denoted by subindexes 1 and 2, respectively. The integration in (B1) is performed along the optical path of the ray.

[53] Evaluation of (B1) is not straightforward and demands extensive derivations, which is beyond the scope of this article. The result is given by the equation below:

$$I_2 = \frac{n_2}{n_1} I_1 e^{-\int_{s_1}^{s_2} \frac{\nabla^2 S}{n} ds} = \frac{r_1^2 \sin(\theta_1) \cos(\psi_1) \left| \frac{d\theta}{da} \right|_1}{r_2^2 \sin(\theta_2) \cos(\psi_2) \left| \frac{d\theta}{da} \right|_2} I_1. \quad (\text{B2})$$

It should be noted that the derivatives are performed at the endpoints.

[54] At the starting point, i.e., the GPS position, $d\theta/da$ and $\sin(\theta)$ are both zero but the intensity is infinite. This means that we must find the limit of $I_1 \sin(\theta_1) (d\theta/da)_1$ when the point S_1 approaches the GNSS position. First, the incremental angle yields:

$$\Delta\theta \cong \frac{1}{r_1} \sin(\psi_1) \Delta s, \quad (\text{B3})$$

where Δs is the distance between the GNSS point and the endpoint r_1 on the ray path. Since the endpoint contains information about the impact parameter, the

derivative $d\theta/da$ at the GNSS point must be found from the integral expression for θ integrated in the r -space:

$$\theta = \int_{r_1}^{r_G} \frac{1}{r} \frac{a}{\sqrt{n^2 r^2 - a^2}} dr, \quad (\text{B4})$$

which yields:

$$\left(\frac{d\theta}{da} \right)_1 \cong \frac{\Delta s}{n_1 r_1^2 \cos^2(\psi_1)}. \quad (\text{B5})$$

If we take the limit of $(\Delta s)^2 I_1 \mu P_G / 2\pi$ for $r_1 \mu r_G$ where P_G is the power transmitted by the GPS satellite, the final expression for the intensity at the LEO receiver becomes:

$$I_L = \frac{P_G}{2\pi} \frac{a_G}{r_G r_L \sin(\theta_L) \sqrt{n_G^2 r_G^2 - a_G^2} \sqrt{n_L^2 r_L^2 - a_L^2}} \left| \frac{d\theta}{da} \right|_L \frac{n_L}{n_G}. \quad (\text{B6})$$

In the calculation of the intensity in (B6) it has not been assumed that the index of refraction is spherical symmetric and it is therefore allowed to have different impact parameters for the two satellites.

[55] When the index of refraction is constant and the ray path is a straight line, (B6) reduces to

$$I_2 = \frac{P_G}{2\pi r_{LG}^2}, \quad (\text{B7})$$

where r_{LG} is the distance between the transmitter and receiver. This is well known and purely a justification of the calculation.

[56] By setting $a_G = a_L = a$ and $n_G = n_L = 1$, the amplitude of the ray at the LEO-receiver can be written as:

$$A_L = \sqrt{\frac{P_G}{2\pi}} \sqrt{\frac{a}{r_G r_L \sin(\theta_L) \sqrt{r_G^2 - a^2} \sqrt{r_L^2 - a^2} \left(\frac{d\theta}{da} \right)_L}}. \quad (\text{B8})$$

In (B8) the absolute value symbol has been removed; it introduces a phase shift of $\pm \pi/2$ depending on the sign of $d\theta/da$. Physically, it resembles the phase anomaly around a focus in an optical system.

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